



Click and Learn
*Using DNA to Trace
Human Migration*

INTRODUCTION

This handout complements the Howard Hughes Medical Institute's Click and Learn "Using DNA to Trace Human Migration." In this activity, you learn how scientists have used genetic markers to trace the migration routes and origins of modern human populations.

PROCEDURE

Follow the instructions as you proceed through the Click and Learn and answer the questions. Please answer the questions in your science notebook or journal.

PART 1: Introduction

1. Consider the following question:

"Scientific evidence suggests that the entire human population descended from just several thousand African migrants about 50,000 years ago." What sort of evidence would you expect to find if this claim were true? List the types of evidence you would expect to find if all humans living today came from a small population in Africa 50,000 years ago.

PART 2: Genetic Markers

2. Go to the Click and Learn found on the BioInteractive website: <http://www.hhmi.org/biointeractive/using-dna-trace-human-migration> and carefully read slides 1–4, ending on "Measuring Genetic Variation."
3. Slide 4 discusses the genetic differences among humans. It also compares genetic similarity between humans and chimpanzees. Why does it make the comparison between humans and chimps? What is the author of the Click and Learn trying to demonstrate?
4. If two populations of people share the same DNA "spelling difference" (i.e., mutation), what does this mean about these two populations?
5. Proceed to reading slide 5. What do geneticists call "spelling differences" that are useful for identifying human populations that share a common ancestor?
6. Read slides 6–7. All people living outside of Africa share a common genetic marker. That marker is not present in most Africans. Could this finding be used as evidence that humans originated in Africa? Why or why not?

PART 3: Out of Africa

7. Read slides 8–11. When does the Click and Learn suggest that humans left Africa for the first time?
8. What locations became populated by humans relatively soon after they left Africa?
9. Read slides 12–14. After humans populated Central Asia, in what other directions did they migrate? Did this all happen at the same time?
10. On slide 14, we learn that there are "about 5,000 extended human families." In your own words, explain what this claim means.



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11. Why would language families be a useful indicator of human migration patterns?
12. Read slide 15 and consider the map and question marks on the slide. What evidence would you look for today to determine which of those routes the original migrants took to Australia?
13. Read from slide 16 to the end. What is the significance of finding the Australian marker in an isolated village in India? What does this tell you about the route taken by the original migrants to Australia?
14. Review your answer to Question 1 at the stop of this handout. Does this evidence support the claim that all humans originated in Africa? Why or why not?
15. This Click and Learn presents two types of evidence: biological (genetic markers) and cultural (languages). Do you think one type of evidence is more convincing than the other? Explain your answer.